Chapter VI – Unsupervised Learning

Instituto Superior de Engenharia
Politécnico de Coimbra

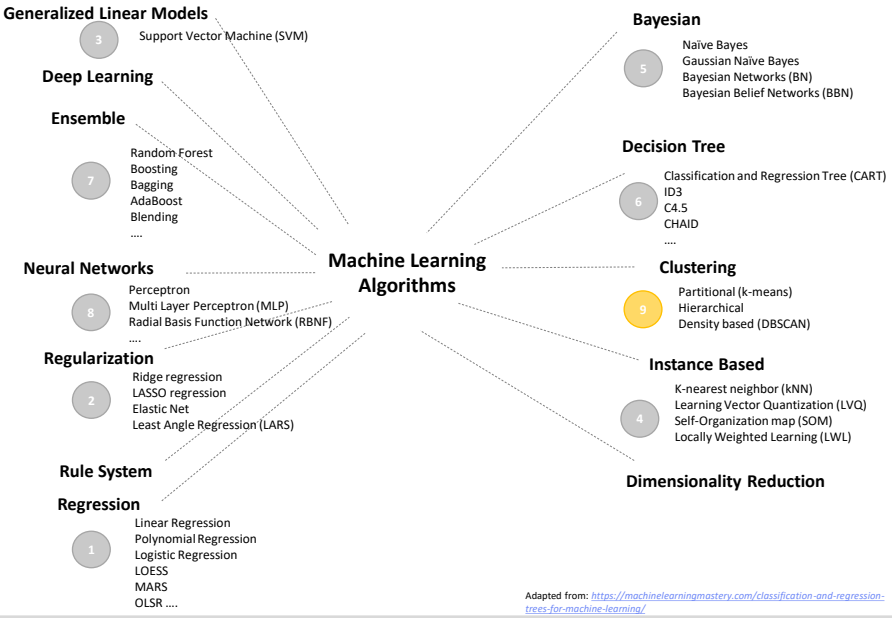
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Outline

- Clustering
  - Partitional Algorithms
  - Hierarchical Algorithms
  - Density Based Algorithms
- Dimensionality Reduction
  - Principal Component Analysis

Clustering

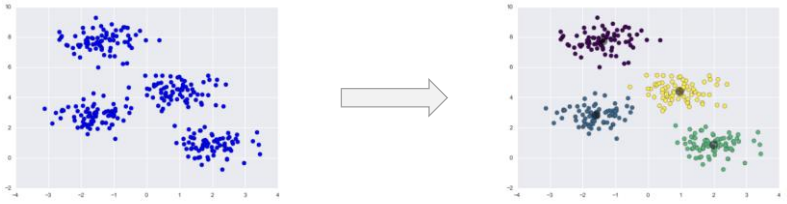
Machine Learning Algorithms



Adapted from: https://machinelearningmastery.com/classification-and-regression-trees-for-machine-learning/

Clustering

- Clustering techniques have a very relevant role in unsupervised learning, where the goal is to find a suitable representation of underlying distribution **of the unlabeled data.**

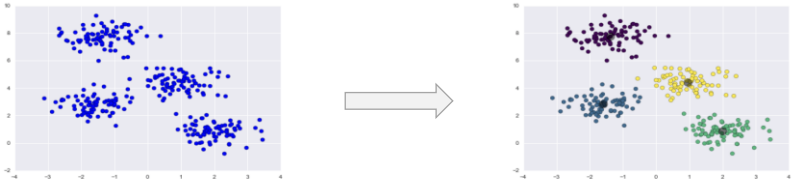


*"Clusters should **reflect some mechanism** at work in the domain from which instances or data points are drawn, that causes **some instances to bear a stronger resemblance to one another than they do to the remaining instances**."*

A Comprehensive Overview of Basic Clustering Techniques, Glenn Fung , University of Wisconsin-Madison.

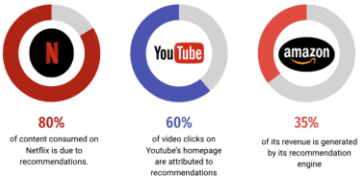
Clustering

- Remember!
  - It is an unsupervised learning problem
    - **There is no label** for training the models
    - The goal is to interpret data (inputs) and group the several instances according to **some pattern contained in the data**.
  - It **cannot be directly** applied to classify instances



Clustering

- It is applied across several real world domains:
  - Recommendation systems



https://code4thought.eu/2022/05/16/fix-your-recommender-system-to-fix-your-profitability/

  - Customer segmentation



https://www.digitalvidya.com/blog/customer-segmentation-for-a-personalized-marketing-experience/

  - …

Clustering

- Clustering
  - Identify similar instances and assigning them to clusters or groups
    - Group instances into clusters such that:
      - $\Upsilon \in R^{N \times p}$ represent a set of N instances $\mathbf{x}_i \in R^p$ with p features
        - The goal is to partition Υ into K groups $G = \{G_1, \ldots, G_K\}$ such that:
          - **Data that belongs to the same group is more similar than data that belongs to other groups.**
      - The concepts of similarity/dissimilarity are based on the resemble between data instances, which can be assessed based on different distance functions (similarity metrics)
        - Selected according to the type of data

Clustering

Features Type		Distance Functions between instances $\mathbf{u} = [u_1 \dots u_p]^T; \mathbf{v} = [v_1 \dots v_p]^T;$ p number of features; r positive integer; ζ number of matches; n_1: 1 *in* $\mathbf{u}, \mathbf{v}$; n_2: 0 *in* $\mathbf{u}, \mathbf{v}$; n_3: 1 *in* $\mathbf{u}$ and 0 in $\mathbf{v}$; n_4: 0 *in* $\mathbf{u}$ and 1 in $\mathbf{v}$; U_k: $\max(u_i)$
Numerical (same Scale)	Euclidean	$d(\mathbf{u}, \mathbf{v}) = \sqrt{\sum_{i=1}^{p} (u_i - v_i)^2}$
	Minkowsky	$d(\mathbf{u}, \mathbf{v}) = \sqrt[r]{\sum_{i=1}^{p} \lvert u_i - v_i \rvert^r}$
	Manhattan	$d(\mathbf{u}, \mathbf{v}) = \sum_{i=1}^{p} \lvert u_i - v_i \rvert$
	Chebychev	$d(\mathbf{u}, \mathbf{v}) = max_{i=1}^{p} \lvert u_i - v_i \rvert$
	Camberra	$d(\mathbf{u}, \mathbf{v}) = \sum_{i=1}^{p} \frac{\lvert u_i - v_i \rvert}{\lvert u_i + v_i \rvert}$
Binary	Simple Matching	$d(\mathbf{u}, \mathbf{v}) = \frac{n_1 + n_2}{p}$
	Jaccard Similarity	$d(\mathbf{u}, \mathbf{v}) = \frac{n_1}{n_1 + n_3 + n_4}$
	Hamming Distance	$d(\mathbf{u}, \mathbf{v}) = n_3 + n_4$
Categorical	Dissimilarity	$d(\mathbf{u}, \mathbf{v}) = \frac{p - \zeta}{p}$
Ordinal	Numerical *	$z_i = \frac{u_i - 1}{U_k - 1}$

* : after the conversion to interval [0,1] distance functions of numerical attributes might be applied

Clustering

- All the distance metrics should verify:

$$d(\mathbf{u}, \mathbf{v}) \geq 0$$

$$d(\mathbf{u}, \mathbf{v}) = 0 \quad if\ and\ only\ if\ \mathbf{u} = \mathbf{v}$$

$$d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u})$$

$$d(\mathbf{u}, \mathbf{z}) \leq d(\mathbf{u}, \mathbf{v}) + d(\mathbf{v}, \mathbf{z})$$

where: $\mathbf{u} = [u_1 \dots u_p]^T; \mathbf{v} = [v_1 \dots v_p]^T; \mathbf{z} = [z_1 \dots z_p]^T$

- The preprocessing of data includes some techniques (encoding, scaling, imputation) to optimize the clustering algorithms operation.
  - However it is also possible to apply a distance metric specific for features of different types (mixed attributes situation)

$$d(\mathbf{u}, \mathbf{v}) = \frac{\sum_{j=1}^{p} \delta_j d_j}{\sum_{j=1}^{p} \delta_j}$$

Clustering

- Mixed Attributes/Features

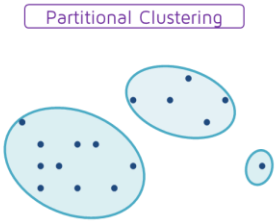
$$d(\mathbf{u}, \mathbf{v}) = \frac{\sum_{j=1}^{p} \delta_j d_j}{\sum_{j=1}^{p} \delta_j}$$

where:

- $\delta_j = 0$ when u_j or v_j is missing, otherwise $\delta_j = 1$
- d_j ?
  - attribute u_j, v_j is **binary** or categorical $d_j = 0$ if $u_j = v_j$ otherwise $d_j = 1$
  - attribute u_j, v_j is **numerical** $d_j = \frac{|u_j - v_j|}{(max_j - min_j)}$, where max_j is the maximum value of attribute j and min_j is the minimum value of attribute j throughout the entire dataset
  - attribute u_j, v_j is **ordinal**, $z_i = \frac{x_i - 1}{max(x_i) - 1}$ must be calculated for u_j, v_j and then the distance for numerical attributes ($d_j = \frac{|u_j - v_j|}{(max_j - min_j)}$) must be applied

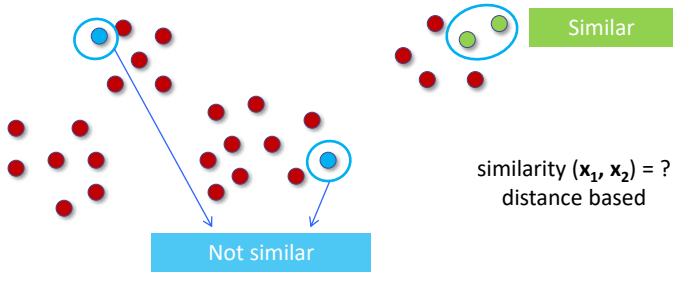
Partitional Algorithms: K-means

Clustering



Clustering

- Partitional clustering
  - Class of algorithms that construct k partitions of the data
    - Each cluster optimizes a clustering criterion
      - such as the minimization for the sum of squared distance from the mean within each cluster.
  - The creation of clusters is based on **similarity** between instances:



Clustering

- Partitional clustering
  - K-means
    - It is a classic partitional algorithm: given $\Upsilon \in R^{N \times p}$ (N instances $\mathbf{x}_i$ with p attributes) may be partitioned into $\boldsymbol{K}$ **groups** G_j , $j = 1, \ldots, K$ that contain the respective **cluster centers** $\boldsymbol{c_j}$ such that:

$$Q_j = \sum_{\substack{r \\ \mathbf{x}_r \epsilon G_j}} \|\mathbf{x}_r - c_j\|^2$$

Q_j is the objective function, i.e. the distance between the instances $\mathbf{x}_r \in G_j$and the respective cluster center c_j.

where:

- K is a priori-defined value (number of clusters)
- Each cluster has at least 1 instance and each instance only belongs to one cluster (most similar cluster center)

Clustering

- The partitioned groups G_j are defined by a $K \times N$ (K clusters and N instances) binary membership matrix.
  - Membership matrix is defined as:

$$u_{ji} = \begin{cases} 1 \;\; if \;\; \|\mathbf{x}_i - c_j\|^2 \leq \|\mathbf{x}_i - c_k\|^2 \;\; , for \; j \neq k \\ 0, \qquad otherwise \end{cases}$$

K-MEANS MEMBERSHIP

Vector	Cluster Centroid	C0	C1	C2	C3
40,4	42.5,4	0	1	0	0
37,4	37.5,3.5	1	0	0	0
45,4	42.5,4	0	1	0	0
34,2	34,2	0	0	0	1
38,5	38,5.5	0	0	1	0
38,6	38,5.5	0	0	1	0
38,3	37.5,3.5	1	0	0	0

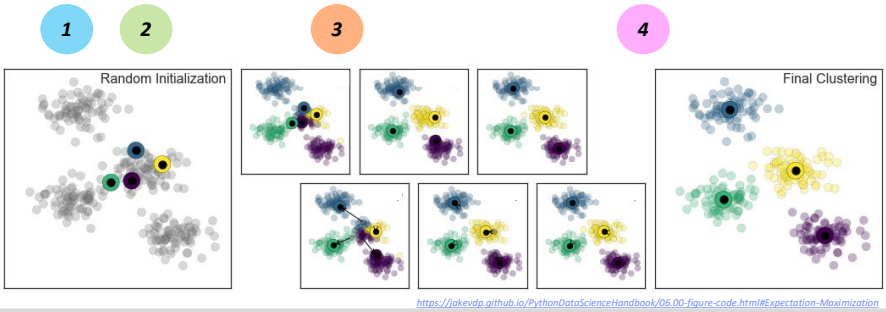
https://www.semanticscholar.org/author/J.-Lu/71008945

- K-means is an iterative method, where the centers of the different clusters can be update as follows:

$$c_j = \frac{1}{|G_j|} \sum_{\substack{r \\ \mathbf{x}_r \epsilon G_j}} \mathbf{x}_r$$

Clustering

- The k-means algorithm can be systematized as:
  - 1. Randomly K cluster centers (K randomly chosen instances) or K randomly defined points are selected.
  - 2. Each instance is assigned to the closest cluster center.
  - 3. The cluster centers are recalculated (updated) using the current cluster memberships.
  - 4. The process should iterate (2 and 3) until a convergence criteria is reached.
    - The centers' values are no longer updated, i.e. the correct value is achieved



https://jakevdp.github.io/PythonDataScienceHandbook/06.00-figure-code.html#Expectation-Maximization

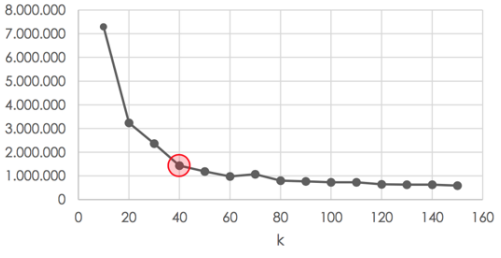
Clustering

- K-means is extensively used in cluster analysis, although it presents several drawbacks:
  - The difficulty in choosing the proper number of clusters centers (k);
  - The performance of the algorithm depends directly on the initialization of the clusters centers that must be a priori defined;
  - The algorithm is sensitive to outliers.
- There are several extensions to original k-means in order to minimize these flaws:
  - k-Medoids
    - The initial k points belong to the dataset
  - Fuzzy clustering
  - …

Clustering

- The proper number of clusters centers?
  - Elbow method
    - run several k-means, incrementing k in each iteration,
    - Compute Q_j for all K clusters

$$Q_j = \sum_{\substack{r \\ \mathbf{x}_r \epsilon G_j}} \|\mathbf{x}_r - c_j\|^2$$



There is a point where the error curve starts to bend (the **elbow point**), that represents an acceptable trade-off between error and number of clusters.

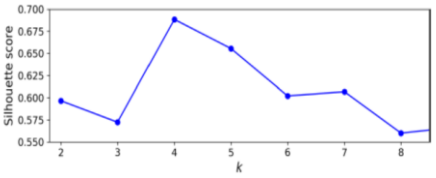
"… it might be hard to determine where the elbow of the curve actually is. The elbow method just gives an orientation where the optimal number of k might be, but it is a very subjective method and for some data sets it might not work…."

https://www.inovex.de/blog/disadvantages-of-k-means-clustering/

Evaluation

- The silhouette coefficient
  - $s(\mathbf{x}_i) = \frac{b(\mathbf{x}_i) - a(\mathbf{x}_i)}{max(a(\mathbf{x}_i), b(\mathbf{x}_i))}$

a: mean **intra cluster** distance
b: mean nearest cluster distance (**inter-cluster**)



Cluster A: A_1, A_2
Cluster B: B_1

Pair Distance	Distance
$d(A_1, A_2)$	2
$d(A_1, B_1)$	8
$d(A_2, B_1)$	7

Silhouette Coefficient for A_1:

$a = d(A_1, A_2) = 2$

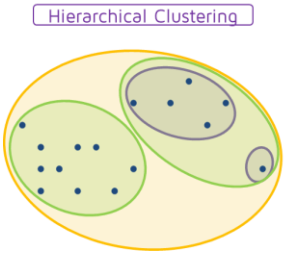
$b = d(A_1, B_1) = 8$

$s = \frac{8-2}{\max(2,8)} = 0.75$

Silhouette ranges from -1 to +1; 0.75 indicates that A_1 is near to instances from its own cluster and far from the other cluster.

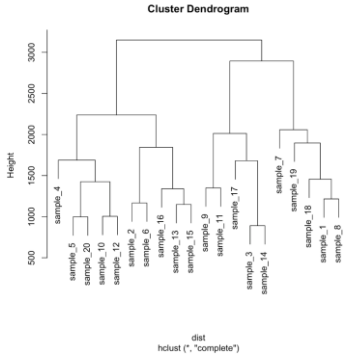
Hierarchical Algorithms

Clustering



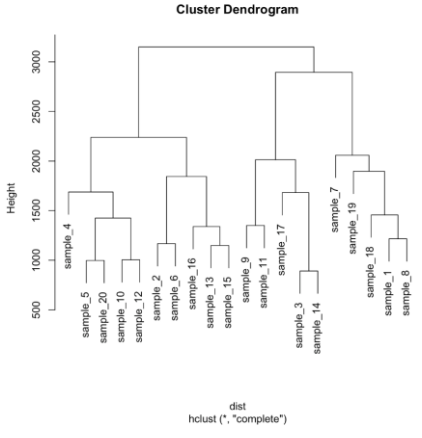
Hierarchical Algorithms

- Hierarchical algorithms create a multilevel hierarchy where clusters at one level are joined as clusters in the next level.
  - Determine cluster assignments by building a hierarchy. This is implemented by either a bottom-up or a top-down approach
  - The graphical representation of this hierarchical decomposition of the data instances is designated as dendrogram



Hierarchical Algorithms

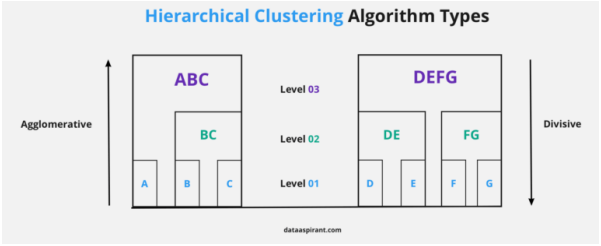
- These methods produce a tree-based hierarchy (dendrogram)



- Clusters are assigned by cutting the dendrogram at a specified depth that results in K groups of smaller dendrograms.

Hierarchical Algorithms

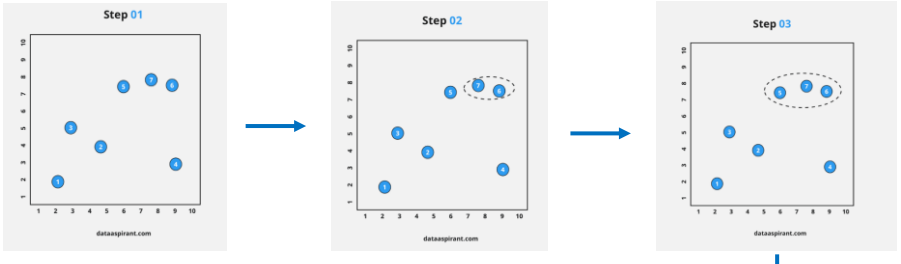
- The methods used to decompose data instances hierarchically can be:
  - Agglomerative (bottom-up)
    - It assumes that each instance is a separate cluster that must be merged according to a specific distance measure.
      - The stop condition is achieved when all the clusters belong to the same cluster.
  - Divisive (top-down).
    - It assumes that all data instances belong to the same cluster which must be split in disjoint clusters.
      - The process iterates until each data instance belongs to a separate cluster or it reaches a stopping condition.

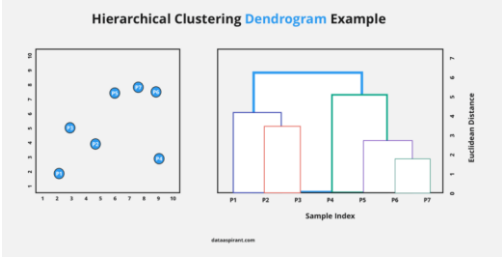


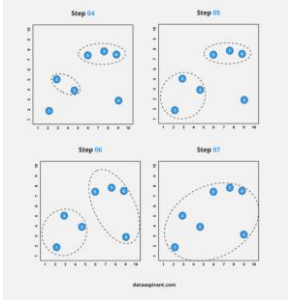
https://dataaspirant.com/hierarchical-clustering-algorithm/

Hierarchical Algorithms

- Agglomerative Clustering



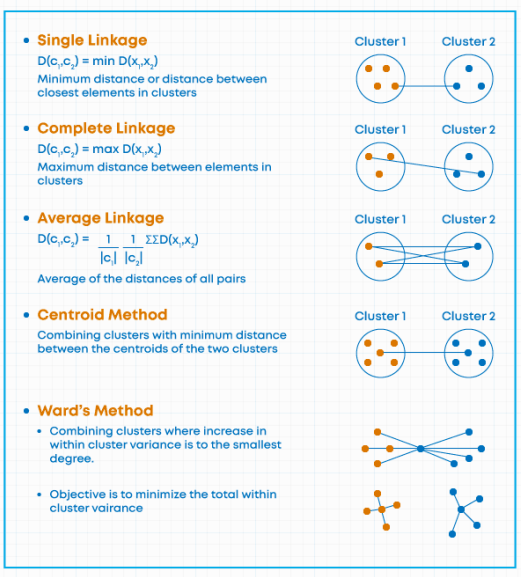




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Hierarchical Algorithms

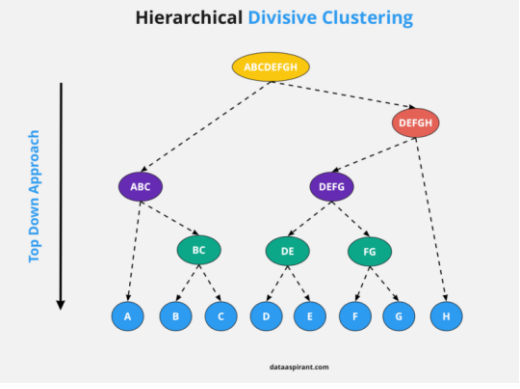
- Nearest Clusters?



https://dataaspirant.com/hierarchical-clustering-algorithm/

Hierarchical Algorithms

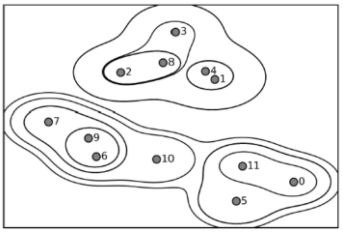
- Divisive Clustering



- All the instances belong to the same cluster.
- Partition the single cluster into the two least similar clusters.
- The process iterates to form the new clusters until the desired number of clusters
  - Limit stop condition is one cluster for each instance.

Hierarchical Algorithms

- Hierarchical Clustering
  - Unlike partitional techniques:
    - An instance belongs to different clusters



Introduction to Machine Learning with Python: A Guide for Data Scientists, O'Reilly, 2016

  - **It is a deterministic process**, meaning cluster assignments won't change when an algorithm is run more than one time on the same input data.

Hierarchical Algorithms

- Hierarchical clustering :
  - Often reveal the finer details about the relationships between data objects.
  - Provide an interpretable dendrogram.
  - However, they are:
    - Computationally expensive
    - Sensitive to noise and outliers.

Density Based Algorithms

Clustering

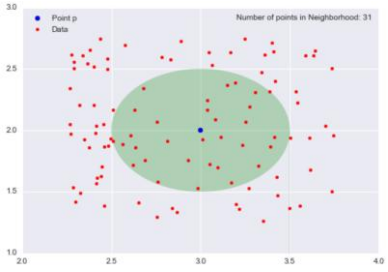
Density Based Algorithms

- Clusters are created based on the instances density
  - Subtractive Clustering Algorithm
  - DBSCAN



Density Based Algorithms

- Subtractive Clustering Algorithm
  - It is a density based algorithm, since it groups data instances according to their density, i.e. the number of data instances in a specific neighborhood.



  - All the data instances are viewed as candidates for cluster centers, being assessed the respective density through the following formula:

$$\rho_{\mathbf{x}_i} = \sum_{j=1}^{N} \exp\left(-\frac{\left\|\mathbf{x}_i - \mathbf{x}_j\right\|^2}{\left(\frac{r_a}{2}\right)^2}\right)$$

r_a is a positive neighborhood radius

Density Based Algorithms

- The first cluster center is the instance $\mathbf{x}_i$ that presents the highest density value of $\rho_{\mathbf{x}_i}$

$$\rho_{\mathbf{x}_i} = \sum_{j=1}^{N} \exp\left(-\frac{\left\|\mathbf{x}_i - \mathbf{x}_j\right\|^2}{\left(\frac{r_a}{2}\right)^2}\right)$$

- The density values are updated:

$$\rho_{\mathbf{x}_i} = \rho_{\mathbf{x}_i} - \rho_{\mathbf{c}_1} \exp\left(-\frac{\left\|\mathbf{x}_i - \mathbf{c}_1\right\|^2}{\left(\frac{r_b}{2}\right)^2}\right)$$

where:

- the parameter $\rho_{\mathbf{c}_1}$ is the density value of the previous cluster center $\mathbf{c}_1$,
- r_b is a positive constant that decreases the density in the neighborhood of $\mathbf{x}_i$.

Density Based Algorithms

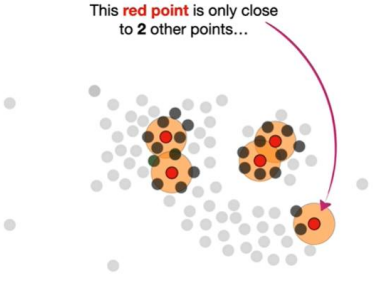
$$\rho_{\mathbf{x}_i} = \rho_{\mathbf{x}_i} - \rho_{\mathbf{c}_1} \exp(-\frac{\|\mathbf{x}_i - \mathbf{c}_1\|^2}{\left(\frac{r_b}{2}\right)^2})$$

- This iterative process forces:
  - The data points near the first cluster center $\mathbf{c}_1$ will have significantly reduced density measure
    - These points become unlikely to be selected as the next cluster center.
  - The constant r_b also defines a neighborhood, usually:

$$r_b = 1.5\, r_a$$

Density Based Algorithms

- DBSCAN - **D**ensity-**B**ased **S**patial **C**lustering of **A**pplications with **N**oise
  - It assumes that clusters are dense regions in space separated by regions of lower density.
    - Densely grouped data points are grouped into a single cluster.

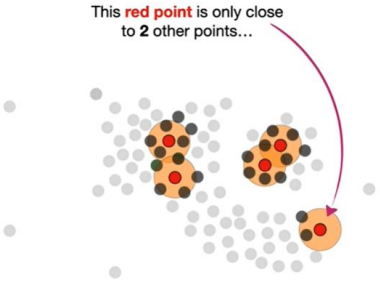


https://www.youtube.com/watch?v=RDZUdRSDOok&t=492s

  - It does not require the initial definition of the number of clusters
  - It is robust to outliers.

Density Based Algorithms

- DBSCAN



  - DBSCAN requires only 2 parameters
    - ε is the radius of the circle to be created around each data point to check the density
    - **minPoints** is the minimum number of data points required inside that circle for that data point to be classified as a **Core point**.

Density Based Algorithms

- DBSCAN
  - A circle of radius ε is created around every data point in order to classify all of them:
    - **Core point**: its circle around contains at least **minPoints (red)**
    - **Border point**: its circle around contains less than **minPoints (yellow)**
    - **Noise:** its circle does not contain any other points around (**purple**)



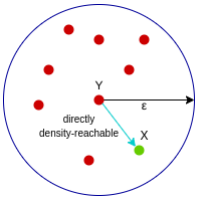
  - Usually, DBSCAN applies Euclidean distance to locate the data instances.

Density Based Algorithms

- DBSCAN
  - Cluster Creation:
    - Two important concepts:
      - Reachability
        - Defines how a data point (instance) can be accessed from another data point
      - Connectivity
        - Defines if two data points belong to the same cluster or not.

Density Based Algorithms

- DBSCAN
  - Cluster creation
    - Points belong to the same cluster or not.
    - Data points might be:
      - Directly Density-Reachable

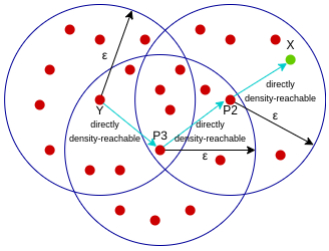


A point $\mathbf{x}$ is directly density-reachable from point $\mathbf{y}$, if:

- $\mathbf{x}$ belongs to the neighborhood of $\mathbf{y}$, i.e, $dist(\mathbf{x}, \mathbf{y}) < \varepsilon$
- $\boldsymbol{y}$ is a core point

Density Based Algorithms

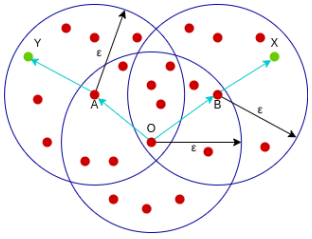
- Density-Reachable



A point $\mathbf{x}$ is density-reachable from point $\mathbf{y}$, if there is a chain of points $\mathbf{p}_i$ where $\mathbf{p}_1 = \mathbf{x}$ and $\mathbf{p}_n = \mathbf{y}$ such that $\mathbf{p}_{i+1}$ is directly density-reachable from $\mathbf{p}_i$

This is a key concept for the cluster formation.

- Density-Connected



A point $\mathbf{x}$ is density-connected from point $\mathbf{y}$, if there is a data points $\mathbf{o}$, such that both $\mathbf{x}$ and $\mathbf{y}$ are density-reachable from $\mathbf{o}$

Cluster creation!

Density Based Algorithms

- Parameters' setup
  - DBSCAN is very sensitive to the values of ε and minPoints.
    - These values change significantly the results of DBSCAN algorithm.
  - **minPoints** should be greater than the number of dimensions of the dataset, i.e. at least:

$$minPoints = p + 1 \quad , p = nr.\ features$$

  - ε can be defined iteratively based on the specific distances:
    - If ε is too small, a high number of clusters will be created and more instances are considered as noise
    - If ε is too high, several clusters will be merged into a big cluster loosing useful information

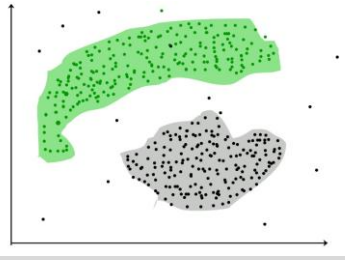
Final Considerations

Clustering

Clusters: Final Considerations

- A clustering algorithm should verify some features:
  - Scalability
    - The ability to perform well with a large number of data instances;
  - Handling of outliers;
  - Insensitivity to the order of attributes in the data instances;
  - Ability to deal with a high number of features

  - Clustering can be applied to:
    - Data Analysis
    - Preprocessing
    - Semi-Supervised Learning
    - …



Evaluation

Clustering

Evaluation

- The evaluation of a cluster algorithm is harder than a supervised learning algorithm.
  - Two different situations might arise:
    - Evaluating clustering **with** ground truth (true data)
      - Adjusted Rand Index (ARI)
        - It is a measure of agreement between two partitions: one given by the clustering process and the other defined by the true data

$$ARI(P^*, P) = \frac{\sum_{i,j}\binom{N_{ij}}{2} - [\sum_i \binom{N_i}{2} \sum_j \binom{N_j}{2}]/\binom{N}{2}}{\frac{1}{2}[\sum_i \binom{N_i}{2} + \sum_j \binom{N_j}{2}] - [\sum_i \binom{N_i}{2} \sum_j \binom{N_j}{2}]/\binom{N}{2}}$$

N is the number of data points in a given data set and N_{ij} is the number of instances of the class $C_j^* \epsilon P^*$ assigned to cluster C_i in partition P.
N_i is the number of data points in cluster C_i, and N_j is the number of data points in class C_j

- $\mathrm{ARI} = 1$ if the cluster is completely identical to the true data.

Evaluation

- Evaluating clustering **with** ground truth (true data)
  - Normalized mutual information (NMI)

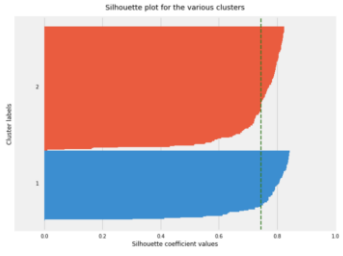
$$NMI(Y, C) = \frac{2 \times I(Y; C)}{H(Y) + H(C)}$$

Y class labels, C cluster labels, $H(\cdot)$ Entropy, $I(Y; C)$ mutual information between Y and C

    - NMI varies between 0 and 1 where it reaches the perfect correlation with true data.
  - Homogeneity
    - Each cluster contains exclusively instances of the same class.
  - Completeness
    - All instances of a given class are assigned to the same cluster.
  - https://scikit-learn.org/stable/modules/clustering.html#clustering-performance-evaluation

Evaluation

- Evaluating clustering **without** ground truth (true data)
  - This is **the most frequent situation**
    - **The silhouette coefficient**



https://medium.com/@cmukesh8688/silhouette-analysis-in-k-means-clustering-cefa9a7ad111

    - Calinski-Harabasz Index
    - Davies-Bouldin Index
    - …

Clustering

- Silhouette coefficient
  - It is a measure of cluster cohesion and separation, as it quantifies how well a data point fits into the respective assigned cluster based on two factors:
    - How close the data point is to other points in the cluster an how far away the data point is from points in other clusters
  - Silhouette coefficient values range between -1 and 1.

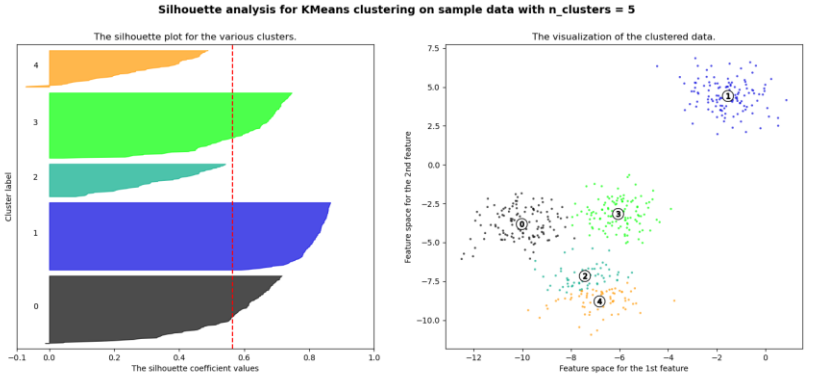
$$s(\mathbf{x}_i) = \frac{b(\mathbf{x}_i) - a(\mathbf{x}_i)}{max(a(\mathbf{x}_i), b(\mathbf{x}_i))}$$

a: mean intra cluster distance
b: mean nearest cluster distance

    - $s(\mathbf{x}_i)$ close to +1 means that the instance is well inside its own cluster and far from other clusters,
    - $s(\mathbf{x}_i)$ close to 0 means that it is close to a cluster boundary
    - $s(\mathbf{x}_i)$ close to –1 means that the instance may have been assigned to the wrong cluster.

Clustering

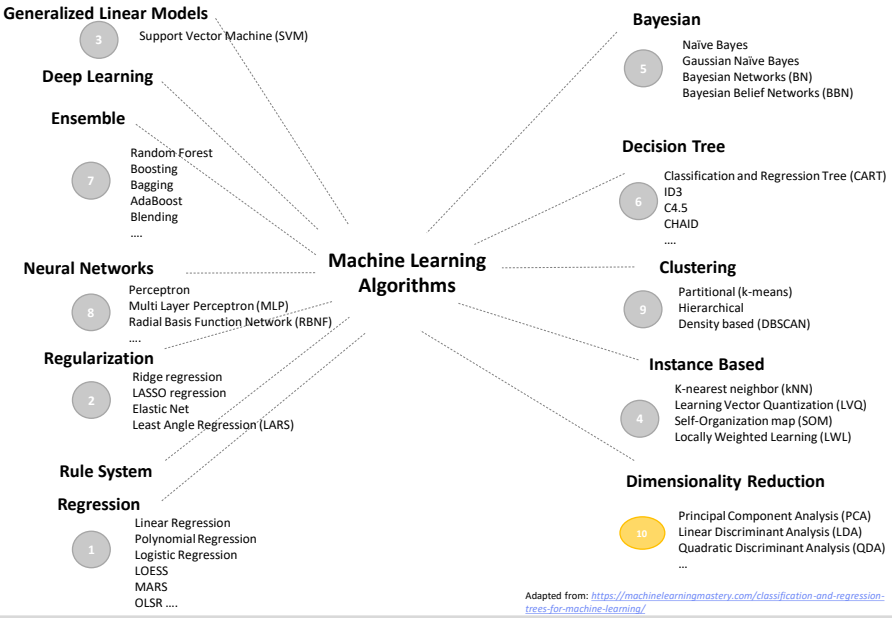
- Silhouette coefficient



https://scikit-learn.org/stable/auto_examples/cluster/plot_kmeans_silhouette_analysis.html

Dimensionality Reduction

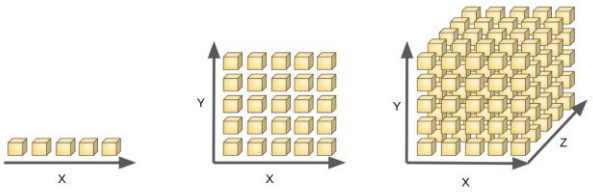
Machine Learning Algorithms



Adapted from: https://machinelearningmastery.com/classification-and-regression-trees-for-machine-learning/

Dimensionality Reduction

- The real world datasets have a great number of features (variables), and consequently high dimensionality (Curse of Dimensionality), which:



  - Decrease the efficiency of algorithms and the respective performance;
  - Increase exponentially the difficulty to compute the data;
  - The functional dependencies in the data become more complex as more attributes contribute to the actual relationships;
  - Concepts like proximity, distance, or neighborhood become less meaningful with the increasing dimensionality of a dataset;
  - Many irrelevant attributes may interfere for instance to find the relevant clusters;
  - In a dataset containing many attributes, there may be some correlations among subsets of attributes.

Dimensionality Reduction

- Dimensionality reduction:
  - Transforms the dataset X with dimensionality D into a new dataset Y with dimensionality d, where $d < D$, and often $d << D$.
  - This is possible as:
    - many variables may have smaller variation than the measurement noise, and thus will not contain relevant information, so they can be discarded;
    - some variables are correlated, thus a set of representative uncorrelated variables can be found, while discarding the remaining ones without a significant loss of information.
      - Intrinsic dimensionality
        - A minimum number of variables that can define the data space without losing its relevant information and properties .
        - Finding this dimensionality, can facilitate the visualization, compression and classification of the data, while avoiding the Curse of Dimensionality.

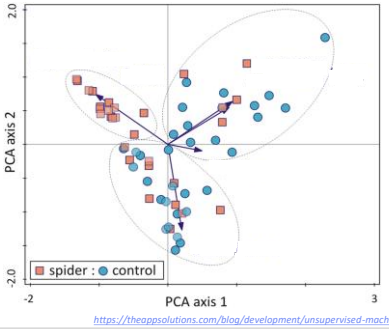
Principal Component Analysis

Dimensionality Reduction

Dimensionality Reduction

- Principal Component Analysis
  - Principal Component Analysis (PCA) is the most representative unsupervised linear technique for dimensionality reduction.
    - It reduces the dimensionality of a large number of correlated variables to a set of new uncorrelated variables, while preserving the maximum variance.
    - This is performed by calculating of the covariance matrix between all dimensions $(D \times D)$ and determining the d principal eigenvalues (principal components).

	Total duration of event*				Delay until first occurrence of event**		
	model	test statistics	d.f.	P	HR	z	P
Penetration event	GLS	F = 21.62	1,61	0.001			
Travel phase	GLS	F = 11.35	1,60	0.001	0.7	−1.985	0.047
Drinking phase							
occurrence†	BGLM‡	z = −1.02		0.309	0.66	−2.173	0.029
duration§	gGLM¶	t = −1.23	58	0.224			
Phloem preparation phase							
occurrence†	BGLM‡	z = −0.64		0.519	0.81	−0.836	0.4
duration§	gGLM¶	t = −2.51	44	0.016			
Phloem ingestion phase							
occurrence†	BGLM‡	z = −2.18		0.029	0.38	−2.082	0.037
duration§	gGLM¶	t = 0.26	24	0.794			



https://theappsolutions.com/blog/development/unsupervised-machine-learning/

END!